

Especificación TLA+ del harness

Máquina de estados de una novela (configuración → planificación → escritura → validación → publicación, con reintentos, crash, reanudación y cambio de un hecho), sus invariantes y propiedades de vivacidad, el resultado de TLC y los contraejemplos que cambiaron el código.

Generado por presentacion/build/build_annexes.py el 2026-09-25 desde el commit c0a9e90. Idioma: español; los ficheros fuente en inglés (código, specs y README técnicos) se reproducen tal cual.

Resumen de la última ejecución de TLC

Fuente: `formal/tla/tlc-output.txt`

Dato	Valor
Resultado	Model checking completed. No error has been found.
Estados generados	10.031.846
Estados distintos	5.492.531
Profundidad	127
Duración	08min 17s
TLC	2026.09.23.154203
Constantes	N = 5, SCENES = 3, MAX_SCENE_RETRIES = 2, MAX_CHAPTER_RETRIES = 2, MAX_REPAIR_ROUNDS = 2, MaxCrashes = 1, MaxChanges = 1

Los cuatro contraejemplos CE1–CE4 (sección 5) aparecieron en borradores anteriores del modelo y cada uno produjo una regla para el código; la tabla acción → código está en la sección 2.

Qué se modela, propiedades y correspondencia con el código

Fuente: `formal/tla/README.md`

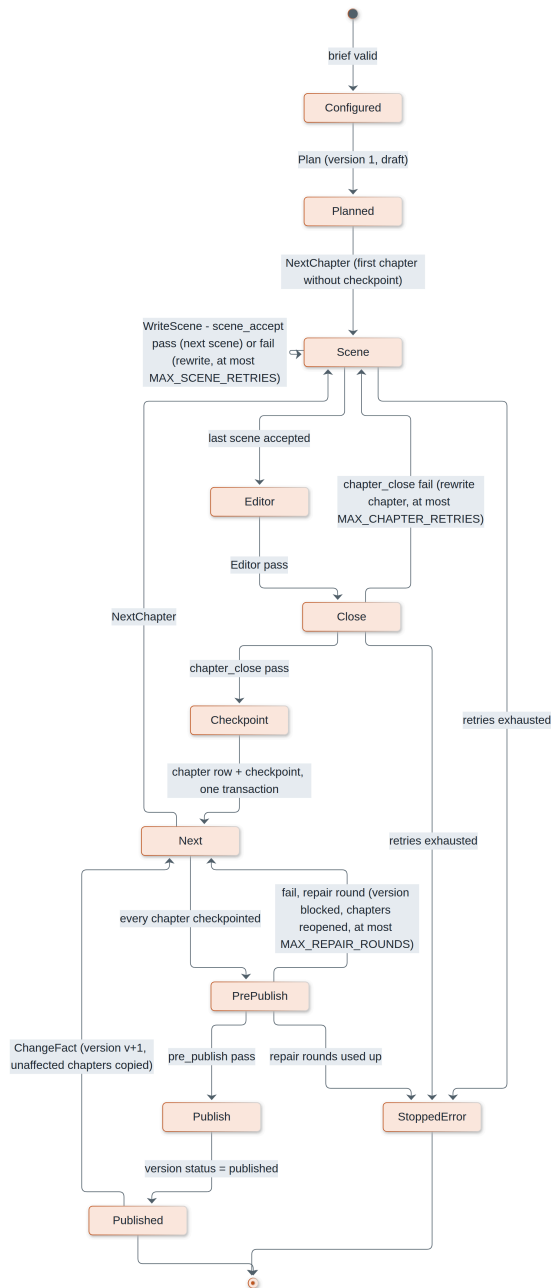
TLA+ model of the gift-novel harness flow

Spec 013 (block B9 of spec 004, exam items T01–T05). Verification method: model checking, **A** (`docs/verification.md`). TLC runs in development, never per generation.

File	What it is
<code>GiftNovelHarness.tla</code>	The model, plain TLA+
<code>GiftNovelHarness.cfg</code>	TLC configuration: constants, invariants, properties
<code>run-tlc.sh</code>	Fetches the pinned <code>tla2tools.jar</code> and runs TLC
<code>tlc-output.txt</code>	Output of the last full run
<code>COUNTEREXAMPLES.md</code>	Every counterexample TLC found while the model was developed, and the code rule each one produced

What is modelled

One novel, from a validated brief to a published version, with crashes and one reader change:



Reading it. Each box is a value of the volatile `pc`. `Crash` is not drawn: it can fire from any state except `Published`, `StoppedError` and `Crashed` itself, wipes the pipeline's memory, and leads to `Crashed`, from which `Resume` reads the database and returns to `Configured` (nothing planned yet), `Published` (latest version already published) or `Next` (a draft or blocked version to finish). `ChangeFact` sends the new version through the same loop: only the chapters it reopened lack a checkpoint, so only they are rewritten.

Durable variables (K1 rows, kept across `Crash`):

Variable	K1 row it stands for
<code>vstatus[v]</code>	<code>novel_version.status</code> $\in$ none (no row), draft, blocked, published
<code>changed[v]</code>	<code>novel_version.changed_chapters</code>
<code>rows[v][c].origin[v][c]</code> , <code>passed[v][c]</code>	<code>chapter_version(version, chapter)</code> : how many rows, in which version the text was written, whether it passed <code>scene_accept</code> and <code>chapter_close</code>
<code>ckpt[v]</code>	<code>checkpoint(version, chapter, status = complete)</code>
<code>ppOk[v]</code>	the persisted <code>pre_publish_validator_result</code> of the version
<code>chFails[v][c]</code>	<code>chapter_close</code> failures in <code>validator_result</code> since the chapter was opened (CE2)
<code>repairRounds[v]</code>	repair rounds used by the version, on <code>novel_version</code> (CE4)
<code>factRev</code>	revision of the brief facts (<code>fact</code>)

Volatile variables (the process's memory, reset by `Crash`): `pc`, `ver`, `cur`, `scene`, `sceneTry`. Scenes are not durable (plan 004, V4): an interrupted chapter restarts from its first scene, so its scene retry counter restarts with it; the budgets that must survive a crash are the chapter and repair counters.

Ghost variables (for the properties only): `crashes`, `changes`, `everPublished`, `snap` (the rows of each version as they were when it was published).

Abstractions. Validator outcomes are nondeterministic (every validator can pass or fail at every call). The set of chapters a reader change affects, and the set a failed `pre_publish` reopens, are nondeterministic non-empty subsets. Prose, prompts, the judge's scores and the MCP server are not modelled. `ChangeFact` (create version, copy unaffected rows and their checkpoints) is one atomic step: the code must do it in one transaction. Concurrent regenerations are out of scope (spec 004).

Properties

Name	Kind	Says
TypeOK	invariant	Every variable stays in its type
NoUnvalidatedPublish	invariant	A published version has a passed <code>pre_publish</code> result, every chapter checkpointed, and every chapter's stored text passed <code>scene_accept</code> and <code>chapter_close</code>
ResumeNoDupNoLoss	invariant	No <code>(version, chapter)</code> is stored twice, and no checkpoint exists without its text
PreviousVersionKept	invariant	Every version ever published is still published and its rows are exactly as they were at publication
RetriesBounded	invariant	Scene retries $\leq$ <code>MAX_SCENE_RETRIES</code> , chapter retries $\leq$ <code>MAX_CHAPTER_RETRIES</code> and repair rounds $\leq$ <code>MAX_REPAIR_ROUNDS</code> , counted across crashes
StoreSurvivesCrash	action property	<code>Crash</code> and <code>Resume</code> change no durable variable, so the set of checkpointed chapters after a resume is exactly the set before the crash (the "no loss" half of resume)
Termination	liveness	Every behaviour ends, and stays, in <code>Published</code> or <code>StoppedError</code>
EveryGenerationEnds	liveness	Whenever the flow is not terminal (a first generation, a resume, a regeneration), it eventually becomes terminal

Liveness holds under **weak fairness** of every step except `Crash` and `ChangeFact` (`WF_vars(Progress)`), with crashes bounded by `MaxCrashes` and reader changes by `MaxChanges`. Without the bounds, a behaviour that crashes forever would never finish, which is true of the real system too.

How to run

```
formal/tla/run-tlc.sh           # needs Java 11+; downloads tools/tla2tools.jar on first run
```

The jar is the v1.8.0 release asset of `tlaplus/tlaplus`, pinned by SHA-256 in the script, because that tag is a rolling pre-release (the pinned jar reports `TLC2 Version 2026.09.23.154203`, rev 4260e47). `tools/` and `TLC's states/` are git-ignored.

Configuration (`GiftNovelHarness.cfg`): `N = 5` chapters, `SCENES = 3` (plan 004, V5), `MAX_SCENE_RETRIES = 2`, `MAX_CHAPTER_RETRIES = 2`, `MAX_REPAIR_ROUNDS = 2` (1 until tuning iteration 1, 2026-09-24; it follows `pipeline.MAX_REPAIR_ROUNDS`), `MaxCrashes = 1`, `MaxChanges = 1`. Nothing had to be reduced.

Last result (`tlc-output.txt`, `MAX_REPAIR_ROUNDS = 2`): *Model checking completed. No error has been found.* 10,031,846 states generated, 5,492,531 distinct, search depth 127, 8 min 17 s on 4 workers. With `MAX_REPAIR_ROUNDS = 1` (the configuration before tuning iteration 1) it was 1,247,479 generated, 696,062 distinct, depth 109, about one minute; no new counterexample appeared with the second round.

Mapping: TLA+ action → code

Confirmed against the code at 340ede7 (spec 007 pipeline, spec 005 repository). Every function below exists under that name; the pipeline is `backend/app/novel/pipeline.py` (`pipeline.`), the repository `backend/app/bible/repository.py` (`repo.`).

TLA+ action	Code	State or transition in the code
Init (pc = "Configured")	app/interview/service.py ingest_brief	a <code>brief</code> row exists and validates (<code>brief_schema</code> , point hook); no <code>novel_version</code> row
Plan	pipeline.plan_novel → persist_plan → record_plan_usage; pipeline._open_version → repo.create_version	one planner call, <code>check_plan</code> + <code>chronology_problems</code> , at most <code>MAX_REPLANS = 1</code> ; the plan is stored as fact <code>plan.v1</code> ; <code>novel_version(version = 1, status = draft)</code>
NextChapter / Resume	pipeline._chapter_loop via <code>repo.first_incomplete_chapter</code> ; entered from <code>generate</code> (<code>resume</code> is the same call)	the lowest chapter of the version without a <code>complete</code> checkpoint; none left → <code>publish_version</code>
WriteScene	pipeline.write_scene → before_scene_accept → run_point(SCENE_ACCEPT)	writer call, then <code>schema_role_output</code> , <code>forbidden_words_scene</code> , <code>no_placeholders</code> ; fail → rewrite with feedback, at most <code>MAX_SCENE_RETRIES = 2</code> (loop counter in memory , see below), then <code>StopRunError</code> → <code>stopped_error</code>
Editor	pipeline.write_chapter → pipeline._edit	the editor joins the three scenes (or repairs a reopened chapter, or rewrites a changed one) plus one bounded length adjustment
CloseChapter	pipeline.close_chapter → before_chapter_close → run_point(CHAPTER_CLOSE)	budget read by <code>_chapter_close_runs</code> from <code>repo.count_chapter_attempts</code> (persisted <code>chapter_close</code> runs since the chapter was last reopened, CE2); fail → the rejected text goes to <code>repo.record_chapter_attempt</code> and the editor rewrites, at most <code>MAX_CHAPTER_RETRIES = 2</code> , then <code>stopped_error</code>
Checkpoint	pipeline.checkpoint → repo.save_chapter_and_checkpoint	one transaction : upsert <code>chapter_version(version, chapter)</code> and <code>checkpoint.status = complete</code> (CE1); refused for a published version (CE3)
PrePublish (pass) / Publish	pipeline.publish_version → before_publish → run_point(PRE_PUBLISH); repo.set_version_status(..., "published")	<code>schema_brief</code> , <code>brief_coverage</code> , <code>lean_chronology</code> , <code>judge_novel</code> , <code>visual_check</code> ; results persisted; the version becomes <code>published</code> and its rows are never written again
PrePublish (fail, repair)	pipeline.publish_version → repo.block_version(..., <code>repair_rounds = repair_rounds + 1</code>) + repo.set_checkpoint(..., "pending")	one transaction: <code>status = blocked</code> , <code>repair_round + 1</code> (CE4), the chapters named by <code>chapters_named</code> reopened; <code>_chapter_loop</code> runs again
PrePublish (fail, exhausted)	pipeline.publish_version → pipeline._stop	<code>repair_rounds >= MAX_REPAIR_ROUNDS = 2</code> : the version stays <code>blocked</code> , the novel <code>stopped_error</code> with <code>stop_reason</code> in the version note
ChangeFact	pipeline.change_fact → repo.update_fact_value → repo.create_version_from(parent, <code>copy_chapters_except = affected</code>) → <code>_chapter_loop</code>	new version v+1 (draft, parent = latest published) with the unaffected chapters and their checkpoints copied in one transaction; only the affected chapters are rewritten; version v untouched
Crash	process killed (container restart, exception, kill)	memory lost; SQLite keeps committed transactions
Resume	pipeline.generate / resume → load_plan, _open_version, first_incomplete_chapter	no version → plan; latest published → "already published"; draft or blocked → continue at the first chapter without a complete checkpoint, with chapter and repair budgets read from the database
Done	—	terminal: <code>RunResult.status ∈ published, stopped_error</code>

Divergences between the model and the code (found while confirming the table; recorded in `docs/process/iteraciones.md`):

1. *Scene retry counter in memory.* `write_scene` counts rewrites in a local loop, so a crash restarts the chapter from scene 1 with a fresh scene budget. This is safe and matches the model, where `sceneTry` is volatile: scenes are never checkpointed (plan 004, V4), and the model's `RetriesBounded` across crashes is enforced by the durable chapter-close and repair counters; the extra cost is bounded by the number of crashes.
2. *ChangeFact is not one transaction in the code.* The model takes it as one atomic step (rule 5 below). `change_fact` commits `update_fact_value` (and, for a name, the renames of cast, derived facts and brief) **before** `create_version_from`, which is itself one transaction. A crash between them leaves the fact changed with no version v+1, and `Resume` then finds the latest version published: the change is lost silently. **Fixed in 772f846**: everything from `update_fact_value` to `create_version_from` and the usage rows runs in one `BibleRepository.transaction()` (savepoint-aware).

Rules the code must follow, each produced by a counterexample (`COUNTEREXAMPLES.md`):

1. The chapter text and its checkpoint are written in one transaction (**CE1**).
2. The chapter retry count comes from persisted `validator_result` rows, not a loop variable (**CE2**).
3. `chapter_version` is keyed by (`version`, `chapter`) and upserted; a published version's rows are never written (**CE3**).
4. The repair round is stored on `novel_version` and updated with the `blocked` status (**CE4**).
5. Assumed by the model, not found by TLC: `change_fact` creates the new version and its copied rows in one transaction, and publishing is an insert of a new version plus a status change, never an update of an earlier version's rows. Both halves hold in the code (the `change_fact` half since 772f846, divergence 2 above).

Especificación TLA+

Fuente: [formal/tla/GiftNovelHarness.tla](#)

```
----- MODULE GiftNovelHarness -----
(* ***** *)
(* Gift-novel harness flow (spec 013, block B9 of spec 004). *)
(* *)
(* One novel: Configured (brief valid) -> Planned -> per chapter: scenes *)
(* with scene_accept and bounded rewrite, editor pass, chapter_close with *)
(* bounded chapter rewrite, Checkpoint(c) -> pre_publish with one bounded *)
(* repair round -> Published(v) or StoppedError. A crash may happen at any *)
(* non-terminal step; Resume restarts from the database. A reader change *)
(* on a published version creates version v+1 and regenerates only the *)
(* affected chapters through the same loop. Validator outcomes are *)
(* nondeterministic. *)
(* *)
(* Durable variables stand for K1 rows (they survive Crash); volatile ones *)
(* are the pipeline's memory (reset by Crash); ghost ones exist only for *)
(* the properties. Counterexamples found while developing the model and *)
(* the fixes they caused are in COUNTEREXAMPLES.md (CE1-CE4 below). *)
(* ***** *)
EXTENDS Naturals, FiniteSets

CONSTANTS N, SCENES, MAX_SCENE_RETRIES, MAX_CHAPTER_RETRIES, MAX_REPAIR_ROUNDS,
  MaxCrashes, MaxChanges

Chapters == 1..N
Versions == 1..(MaxChanges + 1)
Terminal == {"Published", "StoppedError"}
Phases == {"Configured", "Planned", "Next", "Scene", "Editor", "Close",
  "Checkpoint", "PrePublish", "Publish", "Published", "StoppedError",
  "Crashed"}
VStatus == {"none", "draft", "blocked", "published"}

Min(S) == CHOOSE x \in S : \A y \in S : x <= y
Max(S) == CHOOSE x \in S : \A y \in S : x >= y

VARIABLES
  \* durable: rows of the authoritative database (K1); survive a crash
  vstatus, \* novel_version.status
  changed, \* novel_version.changed_chapters
  rows, \* number of chapter_version rows stored per (version, chapter)
  origin, \* version in which the stored text was written (0 = none)
  passed, \* the stored text passed scene_accept and chapter_close
  ckpt, \* checkpoint rows: chapters complete per version
  ppOk, \* persisted pre_publish validator_result passed, per version
  factRev, \* revision of the brief facts
  chFails, \* chapter_close failures in validator_result since the chapter
  \* was last opened (CE2: was an in-memory counter)
  repairRounds, \* pre_publish repair rounds used, on novel_version
  \* (CE4: was an in-memory counter)
  \* volatile: memory of the pipeline process; lost on a crash
  pc, ver, cur, scene, sceneTry,
  \* ghost: bookkeeping for the properties only
  crashes, changes, everPublished, snap

durable == <<vstatus, changed, rows, origin, passed, ckpt, ppOk, factRev, chFails,
  repairRounds>>
volatile == <<pc, ver, cur, scene, sceneTry>>
ghost == <<crashes, changes, everPublished, snap>>
vars == <<durable, volatile, ghost>>

Row(v, c) == <<rows[v][c], origin[v][c], passed[v][c]>>

TypeOK ==
  /\ vstatus \in [Versions -> VStatus]
  /\ changed \in [Versions -> SUBSET Chapters]
  /\ rows \in [Versions -> [Chapters -> Nat]]
  /\ origin \in [Versions -> [Chapters -> 0..(MaxChanges + 1)]]
  /\ passed \in [Versions -> [Chapters -> BOOLEAN]]
  /\ ckpt \in [Versions -> SUBSET Chapters]
  /\ ppOk \in [Versions -> BOOLEAN]
  /\ factRev \in 0..MaxChanges
  /\ pc \in Phases
  /\ ver \in 0..(MaxChanges + 1)
  /\ cur \in 0..N
  /\ scene \in 0..SCENES
  /\ sceneTry \in Nat
  /\ crashes \in 0..MaxCrashes /\ changes \in 0..MaxChanges
  /\ everPublished \subseq Versions
  /\ chFails \in [Versions -> [Chapters -> Nat]]
  /\ repairRounds \in [Versions -> Nat]

Init ==
  /\ vstatus = [v \in Versions |-> "none"]
  /\ changed = [v \in Versions |-> {}]
  /\ rows = [v \in Versions |-> [c \in Chapters |-> 0]]
  /\ origin = [v \in Versions |-> [c \in Chapters |-> 0]]
  /\ passed = [v \in Versions |-> [c \in Chapters |-> FALSE]]
```

```

/\ ckpt    = [v \in Versions |-> {}]
/\ ppOk    = [v \in Versions |-> FALSE]
/\ factRev = 0
/\ pc = "Configured" /\ ver = 0 /\ cur = 0 /\ scene = 0
/\ sceneTry = 0
/\ crashes = 0 /\ changes = 0 /\ everPublished = {}
/\ snap = [v \in Versions |-> [c \in Chapters |-> <<0, 0, FALSE>>]]
/\ chFails = [v \in Versions |-> [c \in Chapters |-> 0]]
/\ repairRounds = [v \in Versions |-> 0]

(* Configured (brief valid) -> Planned: version 1 is created as a draft. *)
Plan ==
/\ pc = "Configured"
/\ vstatus' = [vstatus EXCEPT ![1] = "draft"]
/\ changed' = [changed EXCEPT ![1] = Chapters]
/\ ver' = 1 /\ pc' = "Planned"
/\ UNCHANGED <<rows, origin, passed, ckpt, ppOk, factRev, chFails,
    repairRounds, cur, scene, sceneTry, ghost>>

(* Pick the first chapter of the version without a checkpoint. *)
NextChapter ==
/\ pc \in {"Planned", "Next"}
/\ LET todo == Chapters \ ckpt[ver] IN
    IF todo # {}
    THEN /\ cur' = Min(todo) /\ scene' = 1 /\ sceneTry' = 0
        /\ pc' = "Scene"
    ELSE /\ pc' = "PrePublish"
        /\ UNCHANGED <<cur, scene, sceneTry>>
/\ UNCHANGED <<durable, ver, ghost>>

(* Write one scene and run scene_accept (forbidden words, schema). *)
WriteScene ==
/\ pc = "Scene"
/\ /\ /\ scene < SCENES                                     \* pass, more scenes
    /\ scene' = scene + 1 /\ sceneTry' = 0 /\ UNCHANGED pc
  \/ /\ scene = SCENES                                     \* pass, last scene
    /\ pc' = "Editor" /\ UNCHANGED <<scene, sceneTry>>
  \/ /\ sceneTry < MAX_SCENE_RETRIES                       \* fail, rewrite
    /\ sceneTry' = sceneTry + 1 /\ UNCHANGED <<pc, scene>>
  \/ /\ sceneTry = MAX_SCENE_RETRIES                       \* fail, exhausted
    /\ pc' = "StoppedError" /\ UNCHANGED <<scene, sceneTry>>
/\ UNCHANGED <<durable, ver, cur, ghost>>

Editor ==
/\ pc = "Editor" /\ pc' = "Close"
/\ UNCHANGED <<durable, ver, cur, scene, sceneTry, ghost>>

(* chapter_close: length, exact names, judge. *)
CloseChapter ==
/\ pc = "Close"
/\ /\ /\ pc' = "Checkpoint"                                \* pass
    /\ UNCHANGED <<scene, sceneTry, chFails>>
  \/ /\ chFails[ver][cur] < MAX_CHAPTER_RETRIES           \* fail, rewrite chapter
    /\ chFails' = [chFails EXCEPT ![ver][cur] = @ + 1]
    /\ scene' = 1 /\ sceneTry' = 0 /\ pc' = "Scene"
  \/ /\ chFails[ver][cur] = MAX_CHAPTER_RETRIES           \* fail, exhausted
    /\ pc' = "StoppedError"
    /\ UNCHANGED <<scene, sceneTry, chFails>>
/\ UNCHANGED <<vstatus, changed, rows, origin, passed, ckpt, ppOk, factRev,
    repairRounds, ver, cur, ghost>>

(* Checkpoint(c): the chapter_version row and the checkpoint row are *)
(* written in ONE transaction (CE1). The row is keyed by (version, *)
(* chapter) and upserted, never inserted twice, and only a version that is *)
(* not published may be written (CE3). *)
Checkpoint ==
/\ pc = "Checkpoint"
/\ vstatus[ver] # "published"
/\ rows' = [rows EXCEPT ![ver][cur] = 1]
/\ origin' = [origin EXCEPT ![ver][cur] = ver]
/\ passed' = [passed EXCEPT ![ver][cur] = TRUE]
/\ ckpt' = [ckpt EXCEPT ![ver] = @ \cup {cur}]
/\ chFails' = [chFails EXCEPT ![ver][cur] = 0]
/\ pc' = "Next"
/\ UNCHANGED <<vstatus, changed, ppOk, factRev, repairRounds,
    ver, cur, scene, sceneTry, ghost>>

(* pre_publish: brief coverage, Lean, visual check. *)
PrePublish ==
/\ pc = "PrePublish"
/\ /\ /\ ppOk' = [ppOk EXCEPT ![ver] = TRUE]           \* pass
    /\ pc' = "Publish"
    /\ UNCHANGED <<vstatus, ckpt, chFails, repairRounds>>
  \/ /\ repairRounds[ver] < MAX_REPAIR_ROUNDS             \* fail, repair round
    /\ \E R \in (SUBSET Chapters) \ {} :
        /\ ckpt' = [ckpt EXCEPT ![ver] = @ \ R]
        /\ chFails' = [chFails EXCEPT ![ver] =
            [c \in Chapters |-> IF c \in R THEN 0 ELSE @[c]]]
    /\ vstatus' = [vstatus EXCEPT ![ver] = "blocked"]
    /\ repairRounds' = [repairRounds EXCEPT ![ver] = @ + 1]
    /\ pc' = "Next"
    /\ UNCHANGED ppOk
  \/ /\ repairRounds[ver] = MAX_REPAIR_ROUNDS             \* fail, exhausted

```

```

        /\ vstatus' = [vstatus EXCEPT ![ver] = "blocked"]
        /\ pc' = "StoppedError"
        /\ UNCHANGED <<ppOk, ckpt, chFails, repairRounds>>
    /\ UNCHANGED <<changed, rows, origin, passed, factRev,
        ver, cur, scene, sceneTry, ghost>>

(* Published(version). *)
Publish ==
    /\ pc = "Publish"
    /\ vstatus' = [vstatus EXCEPT ![ver] = "published"]
    /\ everPublished' = everPublished \cup {ver}
    /\ snap' = [snap EXCEPT ![ver] = [c \in Chapters |-> Row(ver, c)]]
    /\ pc' = "Published"
    /\ UNCHANGED <<changed, rows, origin, passed, ckpt, ppOk, factRev,
        chFails, repairRounds, ver, cur, scene, sceneTry,
        crashes, changes>>

(* The reader changes a fact of the published version: version v+1 copies *)
(* the unaffected chapters and regenerates the affected ones. *)
ChangeFact ==
    /\ pc = "Published"
    /\ changes < MaxChanges
    /\ LET w == ver + 1 IN
        /\ w \in Versions
        /\ vstatus[w] = "none"
        /\ \E A \in (SUBSET Chapters) \ {{}} :
            /\ vstatus' = [vstatus EXCEPT ![w] = "draft"]
            /\ changed' = [changed EXCEPT ![w] = A]
            /\ rows' = [rows EXCEPT ![w] = [c \in Chapters |->
                IF c \in A THEN 0 ELSE rows[ver][c]]]
            /\ origin' = [origin EXCEPT ![w] = [c \in Chapters |->
                IF c \in A THEN 0 ELSE origin[ver][c]]]
            /\ passed' = [passed EXCEPT ![w] = [c \in Chapters |->
                IF c \in A THEN FALSE ELSE passed[ver][c]]]
            /\ ckpt' = [ckpt EXCEPT ![w] = Chapters \ A]
        /\ ver' = w
    /\ factRev' = factRev + 1
    /\ changes' = changes + 1
    /\ pc' = "Next"
    /\ UNCHANGED <<ppOk, chFails, repairRounds, cur, scene, sceneTry,
        crashes, everPublished, snap>>

(* The process dies: memory is lost, the database is kept. *)
Crash ==
    /\ pc \notin Terminal \cup {"Crashed"}
    /\ crashes < MaxCrashes
    /\ crashes' = crashes + 1
    /\ pc' = "Crashed" /\ ver' = 0 /\ cur' = 0 /\ scene' = 0
    /\ sceneTry' = 0
    /\ UNCHANGED << durable, changes, everPublished, snap>>

(* Resume from the database: the latest version, first chapter without a *)
(* complete checkpoint. *)
Resume ==
    /\ pc = "Crashed"
    /\ IF \A v \in Versions : vstatus[v] = "none"
        THEN /\ pc' = "Configured" /\ UNCHANGED ver
        ELSE LET v == Max({x \in Versions : vstatus[x] # "none"}) IN
            /\ ver' = v
            /\ pc' = IF vstatus[v] = "published" THEN "Published" ELSE "Next"
    /\ UNCHANGED << durable, cur, scene, sceneTry, ghost>>

Done == pc \in Terminal /\ UNCHANGED vars

Progress == Plan \/ NextChapter \/ WriteScene \/ Editor \/ CloseChapter
    \/ Checkpoint \/ PrePublish \/ Publish \/ Resume

Next == Progress \/ ChangeFact \/ Crash \/ Done

Spec == Init /\ [][Next]_vars /\ WF_vars(Progress)

-----
(* Safety *)

NoUnvalidatedPublish ==
    \A v \in Versions : vstatus[v] = "published" =>
        /\ ppOk[v]
        /\ ckpt[v] = Chapters
        /\ \A c \in Chapters : rows[v][c] >= 1 /\ passed[v][c]

ResumeNoDupNoLoss ==
    \A v \in Versions : \A c \in Chapters :
        /\ rows[v][c] <= 1 \* no chapter stored twice
        /\ c \in ckpt[v] => rows[v][c] = 1 \* no checkpoint without its text

PreviousVersionKept ==
    \A v \in everPublished :
        /\ vstatus[v] = "published"
        /\ \A c \in Chapters : Row(v, c) = snap[v][c]

RetriesBounded ==
    /\ sceneTry <= MAX_SCENE_RETRIES
    /\ \A v \in Versions : repairRounds[v] <= MAX_REPAIR_ROUNDS

```

```
/\ \A v \in Versions : \A c \in Chapters : chFails[v][c] <= MAX_CHAPTER_RETRIES

(* A crash and the resume that follows change nothing in the database.      *)
StoreSurvivesCrash == [][(Crash \ / Resume) => UNCHANGED durable]_vars

(* Liveness *)

Termination == <>[](pc \in Terminal)
EveryGenerationEnds == (pc \notin Terminal) ~> (pc \in Terminal)
=====
```

Configuración de TLC

Fuente: `formal/tla/GiftNovelHarness.cfg`

```
\* TLC configuration for GiftNovelHarness (spec 013, AC 2 and AC 3).
\* Small model required by T03: 5 chapters x 2 retries. MAX_REPAIR_ROUNDS = 2 since tuning iteration 1.
CONSTANTS
  N = 5
  SCENES = 3
  MAX_SCENE_RETRIES = 2
  MAX_CHAPTER_RETRIES = 2
  MAX_REPAIR_ROUNDS = 2
  MaxCrashes = 1
  MaxChanges = 1

SPECIFICATION Spec

INVARIANTS
  TypeOK
  NoUnvalidatedPublish
  ResumeNoDupNoLoss
  PreviousVersionKept
  RetriesBounded

PROPERTIES
  StoreSurvivesCrash
  Termination
  EveryGenerationEnds
```

Salida de TLC (sin las líneas de parseo)

```
TLC2 Version 2026.09.23.154203 (rev: 4260e47)
Running breadth-first search Model-Checking with fp 100 and seed 901940712530085344 with 4 workers on 4 cores with 3042MB heap and 64MB
offheap memory [pid: 19391] (Linux 6.18.44-fc-v37 amd64, Ubuntu 21.0.10 64bit, MSBDiskFPSet, DiskStateQueue).
Starting... (2026-09-24 21:19:10)
Implied-temporal checking--satisfiability problem has 2 branches.
Computing initial states...
Finished computing initial states: 1 distinct state generated at 2026-09-24 21:19:10.
Checking 2 branches of temporal properties for the current state space with 55201 total distinct states at (2026-09-24 21:19:13)
Finished checking temporal properties in 00s at 2026-09-24 21:19:14
Progress(50) at 2026-09-24 21:19:14: 52,535 states generated (52,535 s/min), 29,732 distinct states found (29,732 ds/min), 4,263 states left
on queue.
Checking 2 branches of temporal properties for the current state space with 2083007 total distinct states at (2026-09-24 21:20:14)
Finished checking temporal properties in 20s at 2026-09-24 21:20:34
Progress(79) at 2026-09-24 21:20:34: 1,936,630 states generated (1,884,095 s/min), 1,103,674 distinct states found (1,073,942 ds/min), 124,317
states left on queue.
Progress(86) at 2026-09-24 21:21:34: 4,025,532 states generated (2,088,902 s/min), 2,266,651 distinct states found (1,162,977 ds/min), 184,657
states left on queue.
Checking 2 branches of temporal properties for the current state space with 6611072 total distinct states at (2026-09-24 21:22:34)
Finished checking temporal properties in 01min 04s at 2026-09-24 21:23:38
Progress(93) at 2026-09-24 21:23:38: 6,087,847 states generated (2,062,315 s/min), 3,396,698 distinct states found (1,130,047 ds/min), 182,324
states left on queue.
Progress(100) at 2026-09-24 21:24:38: 8,210,234 states generated (2,122,387 s/min), 4,541,543 distinct states found (1,144,845 ds/min),
137,315 states left on queue.
Progress(127) at 2026-09-24 21:25:38: 10,031,846 states generated, 5,492,531 distinct states found, 0 states left on queue.
Checking 2 branches of temporal properties for the complete state space with 10985062 total distinct states at (2026-09-24 21:25:38)
Finished checking temporal properties in 01min 48s at 2026-09-24 21:27:26
Model checking completed. No error has been found.
  Estimates of the probability that TLC did not check all reachable states
  because two distinct states had the same fingerprint:
  calculated (optimistic): val = 1.4E-6
  based on the actual fingerprints: val = 5.0E-6
10031846 states generated, 5492531 distinct states found, 0 states left on queue.
The depth of the complete state graph search is 127.
The average outdegree of the complete state graph is 1 (minimum is 0, the maximum 31 and the 95th percentile is 2).
Finished in 08min 17s at (2026-09-24 21:27:27)
```

Contraejemplos encontrados

Fuente: `formal/tla/COUNTEREXAMPLES.md`

TLC counterexamples found while developing the model

Spec O13, AC 5 (exam item T05). Every entry below is a real violation reported by TLC on an earlier draft of `GiftNovelHarness.tla`, with the configuration in `GiftNovelHarness.cfg` (`N = 5`, `SCENES = 3`, `MAX_SCENE_RETRIES = 2`, `MAX_CHAPTER_RETRIES = 2`, `MAX_REPAIR_ROUNDS = 1`, `MaxCrashes = 1`, `MaxChanges = 1`). TLC searches breadth-first, so each trace is a shortest one. None was planted: each draft modelled the flow as the K1 schema and plan 004 describe it (`chapter_version` and `checkpoint` are two tables; retry counters are the pipeline's loop variables), and TLC showed where that reading breaks.

Four drafts failed; the fifth passes (see `tlc-output.txt`).

#	Draft	Invariant violated	Trace length	States explored before the violation	Cause in one line
CE1	1	ResumeNoDupNoLoss	18 states	339 distinct	Crash between writing the chapter row and the checkpoint row
CE2	2	RetriesBounded	21 states	549 distinct	Chapter retry counter lived in memory; a crash reset it
CE3	3	ResumeNoDupNoLoss	46 states	3,327 distinct	The repair round re-inserted a chapter row that already existed
CE4	4	RetriesBounded	50 states	7,488 distinct	Repair-round counter lived in memory; a crash reset it

CE1 — a crash between the chapter row and its checkpoint duplicates the chapter

Draft 1. `chapter_close` passing led to two separate actions, `Persist` (insert the `chapter_version` row) and then `Checkpoint` (insert the `checkpoint` row), mirroring the two K1 tables and the two pipeline functions `close_chapter` / `checkpoint`.

Trace (TLC, 18 states).

- Plan — version 1 is a draft.
- NextChapter picks chapter 1; three `WriteScene` steps pass `scene_accept`; `Editor`; `CloseChapter` passes.
- Persist — `rows[1][1] = 1`, but `ckpt[1] = {}`.
- Crash — memory lost, database kept.
- Resume — latest version is 1, a draft; the first chapter without a checkpoint is chapter 1.
- Chapter 1 is written again (3 scenes, editor, close) and `Persist` inserts a second row: `rows[1][1] = 2`.

Violated. `ResumeNoDupNoLoss` (`rows[v][c] <= 1`: no chapter stored twice).

Fix in the model. `Persist` and `Checkpoint` merged into one action `Checkpoint`: the chapter text and its checkpoint are written atomically.

Rule for the code. `checkpoint(version, chapter)` writes the `chapter_version` row and the `checkpoint` row in the same SQLite transaction (`BibleRepository`, one with `conn: block`). A chapter is either fully checkpointed or absent after a crash; `resume` never sees text without a checkpoint.

CE2 — a crash resets the chapter retry budget

Draft 2 (CE1 fixed). The chapter-rewrite counter `chTry` was a volatile loop variable of `write_chapter`, reset when the chapter starts. A ghost `chFails[v][c]` counted `chapter_close` failures since the chapter was last opened, independently of memory.

Trace (TLC, 21 states).

- Plan; chapter 1: three scenes, editor, `CloseChapter` fails → `chTry = 1`, `chFails[1][1] = 1`, rewrite.
- Crash → `chTry = 0` (memory), `chFails[1][1] = 1` (what really happened).
- Resume; chapter 1 restarts with a fresh budget.
- Two more `CloseChapter` failures → `chTry = 2`, `chFails[1][1] = 3`.

Violated. `RetriesBounded` (`chFails[v][c] <= MAX_CHAPTER_RETRIES`): chapter 1 was rewritten three times against a limit of two, and with more crashes the budget is unbounded (each crash buys `MAX_CHAPTER_RETRIES` more model calls).

Fix in the model. The volatile `chTry` is removed; the rewrite decision reads the durable `chFails[ver][cur]`, which `Crash` does not touch.

Rule for the code. The chapter retry count is derived from persisted `validator_result` rows (point `chapter_close`, this version and chapter, since the chapter was last opened), not from a loop variable. `close_chapter` reads it before deciding rewrite vs. `StoppedError`. The count resets when the chapter is checkpointed and when a repair round reopens it.

CE3 — the repair round inserts a second row for a chapter

Draft 3 (CE1, CE2 fixed). A failed `pre_publish` marks the version `blocked`, removes the checkpoints of a non-empty set `R` of chapters and sends them back through the same chapter loop, which ends in the same `Checkpoint` action. `Checkpoint` still *inserted* the row.

Trace (TLC, 46 states).

1. Plan; chapters 1-5 each pass and are checkpointed: `rows[1] = <<1,1,1,1,1>>`, `ckpt[1] = 1..5`.
2. PrePublish **fails**, repair round 1 with `R = {2}`: `vstatus[1] = "blocked"`, `ckpt[1] = {1,3,4,5}`.
3. NextChapter picks chapter 2; it is rewritten and passes; Checkpoint inserts: `rows[1][2] = 2`.

Violated. `ResumeNoDupNoLoss` again, this time with no crash at all: the repair path reuses the insert.

Fix in the model. Checkpoint sets `rows[ver][cur] = 1` (upsert on the key `(version, chapter)`) and is enabled only when `vstatus[ver] # "published"`.

Rule for the code. `chapter_version` has primary key `(version, chapter)`, and checkpoint upserts it (`INSERT ... ON CONFLICT(version, chapter) DO UPDATE`) only while the version is `draft` or `blocked`; the repository refuses any write to a chapter of a `published` version. If spec 005 wants to keep the rejected draft text too (decision D6, "nothing is deleted"), it adds an `attempt` column and a "current" flag; the model's invariant then reads "exactly one current row per `(version, chapter)`". Either way a repair never leaves two current texts for the same chapter.

CE4 — a crash resets the repair-round budget

Draft 4 (CE1-CE3 fixed). The repair counter `repairs` was volatile, like `chTry` in CE2. A ghost `repairTotal[v]` counted repair rounds per version.

Trace (TLC, 50 states).

1. Chapters 1-5 checkpointed; PrePublish fails; repair round: `repairs = 1`, `repairTotal[1] = 1`, one chapter reopened.
2. The reopened chapter is rewritten and checkpointed.
3. Crash → `repairs = 0`; Resume → version 1 is `blocked`, every chapter checkpointed, so NextChapter goes to PrePublish.
4. PrePublish fails again and, seeing `repairs = 0`, opens a **second** repair round: `repairTotal[1] = 2`.

Violated. `RetriesBounded (repairRounds[v] <= MAX_REPAIR_ROUNDS)`: the spec allows one repair round before `StoppedError`.

Fix in the model. The volatile counter is removed; the durable `repairRounds[v]` is incremented in the same step that marks the version `blocked` and reopens the chapters.

Rule for the code. `novel_version` stores the repair round (`repair_round` column, or the count of `pre_publish` failures in `validator_result` for the version), updated in the same transaction that sets `status = 'blocked'` and resets the reopened chapters' `checkpoint.status`. `publish_version` reads it to choose repair vs. `StoppedError`.

Draft 5 — no error

With the four fixes, TLC explores the full state space with no invariant or property violated: 1,247,479 states generated, 696,062 distinct, depth 109, about one minute on 4 workers. The run is saved in `tlc-output.txt`.

Not counterexamples, but checks that the properties are not vacuous (run on scratch copies, not kept in the repository):

- Reachability: TLC reports states where version 2 is published after a crash and a repair round, where version 1 is published after a crash and a repair, and where a generation stops with an error after its repair round. So the invariants are checked on the paths that matter, not on an empty set.
- Mutation M1 — `ChangeFact` regenerates *in place* (writes version v instead of creating v+1): TLC reports `PreviousVersionKept` violated after 954 distinct states.
- Mutation M2 — NextChapter jumps to `Publish` without `PrePublish`: TLC reports `NoUnvalidatedPublish` violated after 657 distinct states.